Introduction

Diabetes is a common chronic disease, and early identification of at-risk individuals is important for timely intervention. This report investigates suitable classification methods for predicting diabetes status using a dataset of 100,000 survey responses. The response variable is diabetes status, and the candidate predictors are gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level and blood glucose level.

The analysis is carried out in three stages. First, exploratory data analysis is used to understand the variables and examine the association between diabetes and each predictor. Second, three classification methods are considered: K-Nearest Neighbours (KNN), Decision Tree (DT) and Logistic Regression (LR). The best version of each method is selected using a random sub-sample of 10,000 observations and an 80:20 train-test split, with accuracy as the main model selection criterion. Finally, the chosen version of each classifier is fitted on the full dataset and compared using accuracy, true positive rate (TPR), ROC and AUC to determine the most suitable method for predicting diabetes status.

Part I: Exploratory Data Analysis

The response variable in this study is diabetes, which is a binary categorical variable taking values 0 and 1, where 1 indicates the presence of diabetes. The predictors include both categorical and quantitative variables. Gender, hypertension, heart disease and smoking history are categorical variables, while age, BMI, HbA1c level and blood glucose level are quantitative variables.

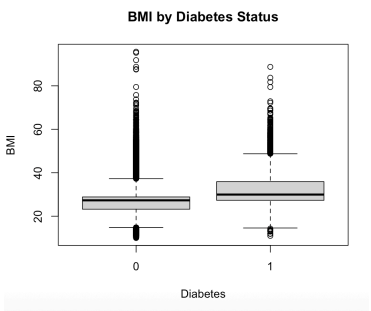
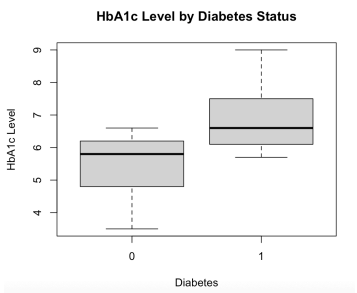
Association between diabetes and categorical predictors

Variables	Rates	Analysis
Diabetes and gender	Rate of having diabetes and male - 0.09748974 Rate of having diabetes and female - 0.07618869	For gender, the difference in diabetes proportions across categories was smaller compared to hypertension and heart disease. Hence, gender appears to have a weaker association with diabetes.
Diabetes and heart_disease	Rate of having diabetes and heart disease - 0.32141045 Rate of having diabetes but no heart disease - 0.07529826	The difference between both rates are relatively significant, hence diabetes and heart disease is said to have a relatively strong association.
Diabetes and hypertension	Rate of having diabetes and hypertension - 0.27895792 Rate of having diabetes but no	For hypertension, the proportion of diabetic individuals was noticeably higher among those with hypertension than among those without

	hypertension - 0.06930768	hypertension. This suggests a relatively strong association between hypertension and diabetes.
Diabetes and smoking_history	Rate of having diabetes and current - 0.10208917 Rate of having diabetes and ever - 0.11788212 Rate of having diabetes and former - 0.17001711 Rate of having diabetes and never - 0.09534122 Rate of having diabetes and No Info - 0.04059638 Rate of having diabetes and not current - 0.10702652	For smoking history, given the similarities in the rates of: (current, ever and former) and (never and not current), it was decided to regroup smoking history into 3 categories: Yes = current, ever, former No = never, not current No info = No Info However, overall, the differences are not as pronounced as those for hypertension and heart disease. Thus, smoking history may still contribute some predictive information, but its association appears weaker.

Table 1

Association between diabetes and quantitative predictors

Variables	Box plot and Correlation	Analysis
Diabetes and BMI	 Figure 1 Correlation Coefficient - 0.2143574	The positions of the medians within each boxplot and their interquartile range (IQR) are relatively similar. Both boxplots exhibit a large number of outliers, with the left boxplot containing relatively more outliers. BMI and diabetes have a correlation coefficient of approximately 0.214. Coupled with the mostly overlapping boxplots, this suggests that the association between BMI and diabetes is relatively weak
Diabetes and HbA1c_level	 Figure 2 Correlation Coefficient -	The positions of the medians within each boxplot are moderately different. Both boxplots exhibit similar length of IQR and no outliers. The correlation coefficient of diabetes and Hb level is 0.401. Coupled with the moderately overlapping boxplots, this suggests that the association between Hb level and diabetes is moderately strong.

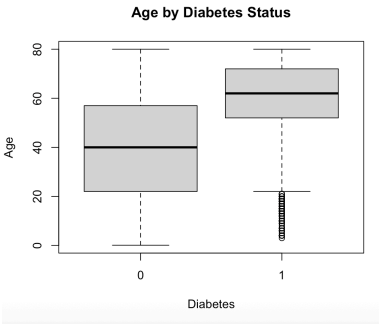
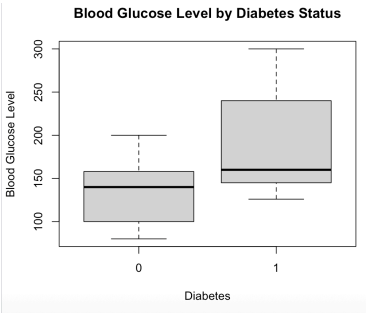
	0.4006603	
Diabetes and age	 Figure 3 Correlation Coefficient - 0.258008	The positions of the medians within each boxplot are quite different. The left boxplot corresponding to individuals without diabetes exhibits larger interquartile range (IQR) compared to the right boxplot that corresponds to individuals with diabetes. The right boxplot has relatively more outliers. The correlation coefficient of age and diabetes is 0.258. Coupled with the mostly overlapping boxplots, this suggests that the association between Age and diabetes is relatively weak.
Diabetes and blood_glucose_level	 Figure 4 Correlation Coefficient - 0.419558	The positions of the medians within each boxplot are moderately different. The correlation coefficient of glucose and diabetes is 0. 420. Coupled with the moderately overlapping boxplots, this suggests that the association between glucose and diabetes is moderately strong

Table 2

Overall, the EDA suggests that HbA1c level and blood glucose level are among the strongest predictors of diabetes status. Hypertension and heart disease also show meaningful associations, while age, BMI, smoking history and gender appear to have weaker but still potentially useful predictive value. Therefore, all variables were considered as candidate inputs in the modelling stage.

Part II: Methods and Model Selection

Three classification methods were considered in this study: KNN, Decision Tree and Logistic Regression. To ensure a fair comparison, the same random sub-sample of 10,000 observations and the same 80:20 train-test split were used in Step 4 for tuning the models. Accuracy was used as the main criterion for selecting the best version of each method. Predicted probabilities for Decision Tree and Logistic Regression were converted to class labels using a threshold of 0.5.

1. K-Nearest Neighbours: The KNN classifier was fitted using all input variables. Since KNN is distance-based, the categorical predictors were converted into dummy variables and all predictors were standardised using the training data before model fitting. This is

necessary because predictors measured on larger numerical scales would otherwise dominate the distance calculations.

To determine the best value of (k), values from 1 to 20 were examined. For each (k), the model was trained on the training set and evaluated on the test set. Accuracy, TPR, FPR, FNR and precision were computed and plotted against (k).

Based on the tuning results, the value k = 9 achieved the highest test accuracy and was therefore selected as the best version of the KNN classifier. The performance curve showed that precision increased with k values, while TPR decreased and FNR increased as well. Thus, k = 9 is a good selection with the highest accuracy, higher precision and a relatively low FNR. The final KNN model for Step 5 was then trained using this selected value of k = 9.

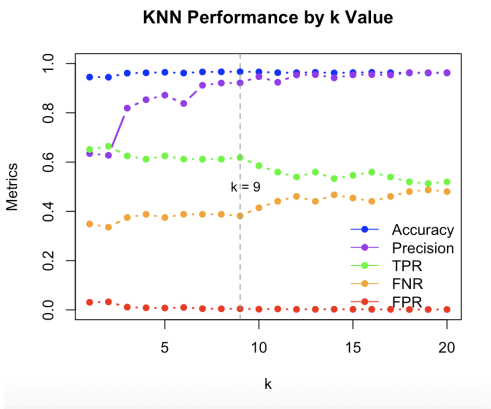


Figure 5

2. Decision Tree: The Decision Tree classifier was fitted using all input variables. A range of complexity parameter (cp) values from 10^-1 to 10^-10 was considered in order to determine the most suitable level of tree complexity. For each (cp) value, a decision tree was fitted on the training set and evaluated on the test set using accuracy, TPR, FPR, FNR and precision.

The value cp = 0.1 produced the highest test accuracy and was therefore chosen as the best version of the Decision Tree classifier. From the tuning results, it was observed that larger cp values tended to reduce predictive accuracy and precision, although TPR increased. At a cp = 0.1, there is a good balance, with highest accuracy and a higher precision. The final Decision Tree model was then fitted using the selected cp = 0.1.

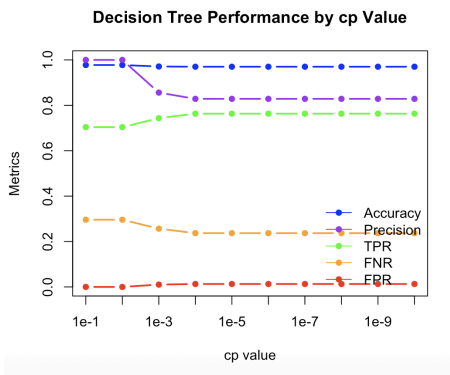


Figure 6

3. Logistic Regression: The Logistic Regression classifier was fitted using the candidate input variables. Since Logistic Regression produces predicted probabilities, a threshold of 0.5 was used to classify observations into diabetic and non-diabetic groups. Several candidate Logistic Regression models were considered, including a full model with all predictors and simpler reduced models, in order to identify a model that was not unnecessarily complex but still achieved strong predictive performance.

The candidate models were compared on the same test set, and accuracy was used as the main selection criterion. The model with the highest test accuracy was selected as the best Logistic Regression classifier. The selected model was full, with all input variables considered, indicating that retaining all predictors gave the strongest predictive performance.

The coefficient summary of the selected Logistic Regression model was also examined. Quantitative predictors such as HbA1c level and blood glucose level were found to be strongly associated with diabetes status, while categorical predictors such as hypertension and heart disease also contributed meaningfully to the model.

Part II: Final Comparison on Full Data

After identifying the best version of each classification method in Step 4, the final comparison was conducted on the full dataset of 100,000 observations. The full dataset was split into an 80:20 training and test set, and the same split was used for KNN, Decision Tree and Logistic Regression to ensure a fair comparison. Each final classifier was trained on the training set using its selected tuning parameter or model form, and its performance was evaluated on the test set using accuracy, TPR, ROC and AUC. The table below summarises the final performances of the three classifiers.

Method	Accuracy	TPR	AUC
KNN	0.96245	0.6094118	0.9314886
Decision Tree	0.97325	0.6852941	0.8426471

Logistic Regression	0.96175	0.6429412	0.9623474

Table 3

The ROC curves of the three classifiers are shown below:

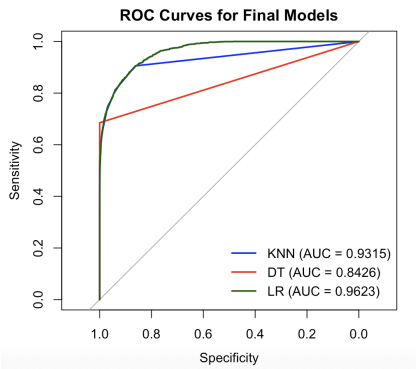


Figure 7

A classifier with a curve closer to the top-left corner and a higher AUC is considered to have better discrimination ability. *From the final comparison, Logistic Regression was selected as the best classifier because it achieved the strongest overall test performance, with the highest AUC and competitive accuracy and TPR. In addition, it is more interpretable than KNN and less unstable than Decision Tree.*

Part III: Conclusion

In this report, three classification methods were investigated for predicting diabetes status: KNN, Decision Tree and Logistic Regression. An exploratory data analysis showed that HbA1c level and blood glucose level were the strongest predictors of diabetes, while age, hypertension and heart disease also displayed meaningful associations. BMI, smoking history and gender appeared to have weaker but still potentially useful relationships with the response.

Using a random sub-sample of 10,000 observations, the best version of each method was selected based on test accuracy. The chosen models were then refitted on the full dataset and compared using accuracy, TPR, ROC and AUC. Based on the final comparison, Logistic Regression was selected as the best classifier for this study.

The main strength of KNN is that it is simple and non-parametric, and can capture local structure in the data. However, it is computationally expensive on large datasets and is sensitive to the choice of k and scaling of predictors. The Decision Tree classifier is easy to interpret and can model non-linear decision boundaries. However, its performance can be unstable, and overly complex trees may overfit the training data. The Logistic Regression classifier is relatively simple, interpretable and computationally efficient. It also provides estimated probabilities directly, which makes ROC and AUC analysis straightforward. However, it assumes a specific functional relationship between the predictors and the log-odds of the response.